

MODULE - 4

Eigen Values & Eigen Vectors of Symmetric matrices

- Jacobi Method for Symmetric matrices

Let A be a given real symmetric matrix. The eigenvalues of A are real and there exist a real orthogonal matrix S such that $S^{-1}AS$ is a diagonal matrix D . The diagonalization is done by applying a series of orthogonal transformations $S_1, S_2, \dots, S_n, \dots$ as follows.

Among the off-diagonal elements, let $|a_{ik}|$ be the numerically largest element. Then, the elements $a_{ii}, a_{ik}, a_{ki}, a_{kk}$ form a 2×2 submatrix A_1 , which can be transformed to a diagonal form. We choose

$$S_1^* = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \text{ and find } \theta \text{ such that the } 2 \times 2 \text{ matrix } A_1 \text{ is diagonalized.}$$

We have,

$$(S_1^*)^{-1} A_1 S_1^* = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} a_{ii} & a_{ik} \\ a_{ki} & a_{kk} \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$\therefore (S_1^*)^{-1} A_1 S_1^* = \begin{bmatrix} a_{ii} \cos^2 \theta + a_{kk} \sin^2 \theta & (a_{kk} - a_{ii}) \sin \theta \cos \theta \\ (a_{kk} - a_{ii}) \sin \theta \cos \theta & a_{ii} \sin^2 \theta + a_{kk} \cos^2 \theta \end{bmatrix}$$

We now choose θ such that this matrix reduces to a diagonal form. That is, we put

$$\frac{1}{2} (a_{kk} - a_{ii}) \sin 2\theta + a_{ik} \cos 2\theta = 0$$

$$\text{i.e. } \tan 2\theta = \frac{2a_{ik}}{a_{ii} - a_{kk}}$$

This equation produces four values of θ and in order that we may get smallest rotation, we require

$$-\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4}, \text{ if } a_{ii} \neq a_{kk} \text{ and}$$

$$\theta = \begin{cases} \pi/4, & a_{ik} > 0 \\ -\pi/4, & a_{ik} < 0 \end{cases} \text{ if } a_{ii} = a_{kk}$$

With the value of θ , the off-diagonal elements vanish and the diagonal elements are simplified. The first step is now completed by performing the rotation $S_1^{-1} A S_1$. In the next step, the largest off-diagonal element in magnitude in the new rotated matrix is found and the procedure is repeated.

We now perform a series of such two dimensional rotations. After finding θ at each step, the rotation is performed with the corresponding orthogonal matrix. For example, if $|a_{ik}|$ is the largest off-diagonal element then we write S_1 as :-

$$S_1 = \begin{bmatrix} 1 & & & & \\ & 1 & & & \\ & & \ddots & & \\ & & & \cos\theta & -\sin\theta \\ & & & \vdots & \vdots \\ & & & +\sin\theta & \cos\theta \\ & & & & \ddots & 1 \end{bmatrix}$$

where $\cos\theta$, $-\sin\theta$, $\sin\theta$, $\cos\theta$ are located in (i, i) , (i, k) , (k, i) and (k, k) positions respectively.

After making ' s ' transformations, we get

$$\begin{aligned} B_s &= S_s^{-1} S_{s-1}^{-1} \dots S_1^{-1} A S_1 \dots S_{s-1} S_s \\ &= (S_1 S_2 \dots S_s)^{-1} A (S_1 \dots S_{s-1} S_s) \\ &= S^{-1} A S \end{aligned}$$

where $S = S_1 S_2 \dots S_s$

As $s \rightarrow \infty$, B_s approaches a diagonal matrix with the eigenvalues on the leading diagonal. We then have the eigenvectors as the corresponding columns of S .

① Using the Jacobi method, find all the eigenvalues and the corresponding eigenvectors of the matrix

$$A = \begin{bmatrix} 1 & \sqrt{2} & 2 \\ \sqrt{2} & 3 & \sqrt{2} \\ 2 & \sqrt{2} & 1 \end{bmatrix}$$

Solⁿ: The largest off-diagonal element is $a_{13} = a_{31} = 2$.
The other two elements in this 2×2 submatrix are $a_{11} = 1$ and $a_{33} = 1$.

$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{4}{0} \right) = \frac{\pi}{4}$$

$$S_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

The first rotation gives $B_1 = S_1^{-1} A S_1$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & \sqrt{2} & 2 \\ \sqrt{2} & 3 & \sqrt{2} \\ 2 & \sqrt{2} & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 2 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

The largest off-diagonal element in magnitude in B_1 is $a_{12} = a_{21} = 2$. The other elements are $a_{11} = 3, a_{22} = 3$

$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{4}{0} \right) = \frac{\pi}{4}$$

$$S_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The second rotation gives $B_2 = S_2^{-1} B_1 S_2 = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

We have the matrix of eigenvectors as $S = S_1 S_2$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

The eigenvalues are 5, 1, -1 and the corresponding eigenvectors are the columns of S .

② Find all the eigenvalues of the matrix

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 2 \\ -1 & 2 & 1 \end{bmatrix}$$

using the Jacobi method. Iterate till the off-diagonal elements in magnitude are less than 0.0005.

Soln:- The largest off-diagonal element in magnitude is a_{12} . Therefore, $\tan 2\theta = \frac{2a_{12}}{a_{11} - a_{22}} = \frac{4}{0} = \infty$

i.e. $\theta = \frac{\pi}{4}$. $\therefore S_1 = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$

The first rotation gives $B_1 = S_1^{-1} A S_1 = \begin{bmatrix} 3 & 0 & 1/\sqrt{2} \\ 0 & -1 & 3/\sqrt{2} \\ 1/\sqrt{2} & 3/\sqrt{2} & 1 \end{bmatrix}$

The largest off-diagonal element in magnitude in B_1 is a_{23} . Therefore, $\tan 2\theta = \frac{2a_{23}}{a_{22} - a_{33}} = -2.121320$

i.e. $\theta = -0.565143$.

$$\cos \theta = 0.844512; \sin \theta = -0.535537$$

$$S_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

The second rotation gives $B_2 = S_2^{-1} B_1 S_2$

$$= \begin{bmatrix} 3 & -0.378682 & 0.597160 \\ -0.378682 & -2.345209 & -0.000003 \\ 0.597160 & -0.000003 & 2.345209 \end{bmatrix}$$

The largest off-diagonal element in magnitude in B_2 is a_{13} . Therefore,

$$\tan 2\theta = \frac{2a_{13}}{a_{11} - a_{33}} = 1.823969 \text{ i.e. } \theta = 0.534647$$

$$\cos\theta = 0.860449 ; \sin\theta = 0.509537$$

$$S_3 = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$$

The third rotation gives $B_3 = S_3^{-1} B_2 S_3$

$$= \begin{bmatrix} 3.353625 & -0.325837 & 0.000001 \\ -0.325837 & -2.345208 & 0.192952 \\ 0.000001 & 0.192952 & 1.991585 \end{bmatrix}$$

The largest off-diagonal element in magnitude in B_3 is a_{12} . Therefore, $\tan 2\theta = \frac{2a_{12}}{a_{11} - a_{22}} = -0.114352$

i.e. $\theta = -0.056929$.

$\therefore \cos\theta = 0.998380$; $\sin\theta = -0.056898$

$$S_4 = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The fourth rotation gives $B_4 = S_4^{-1} B_3 S_4$

$$= \begin{bmatrix} 3.372195 & 0 & -0.010979 \\ 0 & -2.363717 & 0.192639 \\ -0.010979 & 0.192639 & 1.991585 \end{bmatrix}$$

The largest off-diagonal element in magnitude in B_4 is a_{23} . Therefore, $\tan 2\theta = \frac{2a_{23}}{a_{22} - a_{33}} = -0.088462$

i.e. $\theta = -0.044116$.

$\therefore \cos\theta = 0.999027$; $\sin\theta = -0.044102$

$$S_5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

The fifth rotation gives $B_5 = S_5^{-1} B_4 S_5$

$$= \begin{bmatrix} 3.372195 & 0.000484 & -0.010968 \\ 0.000484 & -2.372281 & -0.000003 \\ -0.010968 & -0.000003 & 2.000089 \end{bmatrix}$$

The largest off-diagonal element in magnitude in B_5 is a_{13} . Therefore, $\tan 2\theta = \frac{2a_{13}}{a_{11} - a_{33}} = -0.015987$

i.e $\theta = -0.007992$

$\therefore \cos \theta = 0.999968$; $\sin \theta = -0.007992$

$$S_6 = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}$$

The sixth rotation gives $B_6 = S_6^{-1} B_5 S_6$

$$= \begin{bmatrix} 3.372283 & 0.000484 & 0 \\ 0.000484 & -2.372281 & 0 \\ 0 & 0 & 1.999998 \end{bmatrix}$$

Hence, the approximate eigenvalues are

3.372283 , -2.372281 , and 1.999998 .

==

- Given's method for Symmetric matrices.

The Given method is used to find eigenvalues of a real symmetric matrices $A = [a_{ij}]$. This method preserves the zeros in the off-diagonal elements, once they are created. This method works into two steps. In the first step, the given symmetric matrix is converted to a symmetric tri-diagonal matrix using plane rotations. By a tridiagonal matrix, we mean a matrix of

the form $A = \begin{bmatrix} a_1 & b_2 & 0 & 0 & \dots & 0 & 0 \\ b_2 & a_2 & b_3 & 0 & \dots & 0 & 0 \\ 0 & b_3 & a_3 & b_4 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & b_n & a_n \end{bmatrix}$

The characteristic equation of this matrix is :

$$P_n(\lambda) = \begin{vmatrix} a_1 - \lambda & b_2 & 0 & 0 & \dots & 0 & 0 \\ b_2 & a_2 - \lambda & b_3 & 0 & \dots & 0 & 0 \\ 0 & b_3 & a_3 - \lambda & b_4 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & b_n & a_n - \lambda \end{vmatrix} = 0$$

Expanding by minors, the sequence $\{P_n(\lambda)\}$ satisfies the following equations:

$$P_0(\lambda) = 1$$

$$P_1(\lambda) = a_1 - \lambda$$

$$P_{k+1}(\lambda) = (a_{k+1} - \lambda)P_k(\lambda) - b_{k+1}^2 P_{k-1}(\lambda)$$

$$k = 1, 2, \dots, n.$$

The polynomial $P_n(\lambda)$ is the characteristic polynomial of A .

In second step, the eigenvalues of this new matrix are determined by the method of Jacobi. i.e the conversion to a tri-diagonal form is done by using orthogonal transformations as in Jacobi's method. In this case, the rotation is performed with the elements a_{22} , a_{23} , a_{32} and a_{33} .

$$\text{and } \tan \theta = \frac{a_{13}}{a_{12}}.$$

① Find the eigenvalues of the symmetric matrix

$$A = \begin{bmatrix} 2 & 3 & -1 \\ 3 & 1 & 2 \\ -1 & 2 & -1 \end{bmatrix} \text{ using Givens method.}$$

Soln:- let the orthogonal matrix S_1 be

$$S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

$$\text{where } \tan\theta = \frac{a_{13}}{a_{12}} = -\frac{1}{3} \text{ i.e. } \theta = -0.32175$$

$$\text{Therefore, } S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.94868 & 0.31623 \\ 0 & -0.31623 & 0.94868 \end{bmatrix}$$

The reduced matrix is $B = S_1^{-1} A S_1$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.94868 & -0.31623 \\ 0 & 0.31623 & 0.94868 \end{bmatrix} \begin{bmatrix} 2 & 3 & -1 \\ 3 & 1 & 2 \\ -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.9486 & 0.3162 \\ 0 & -0.3162 & 0.9486 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 3.16228 & 0.00001 \\ 3.16228 & -0.40001 & 2.2 \\ 0.00001 & 2.2 & 0.40001 \end{bmatrix}$$

$$\text{Let } B = \begin{bmatrix} 2 & 3.1623 & 0 \\ 3.1623 & -0.4 & 2.2 \\ 0 & 2.2 & 0.4 \end{bmatrix}$$

The Sturm sequence is :-

$$P_0(\lambda) = 1$$

$$P_1(\lambda) = 2 - \lambda$$

$$\begin{aligned} P_2(\lambda) &= (-0.4 - \lambda) P_1(\lambda) - (3.1623)^2 P_0(\lambda) \\ &= (-0.4 - \lambda)(2 - \lambda) - (3.1623)^2 (1) \\ &= \lambda^2 - 1.6\lambda - 10.8 \end{aligned}$$

$$\begin{aligned} P_3(\lambda) &= (0.4 - \lambda) P_2(\lambda) - (2.2)^2 P_1(\lambda) \\ &= (0.4 - \lambda)(\lambda^2 - 1.6\lambda - 10.8) - (2.2)^2 (2 - \lambda) \\ &= -\lambda^3 + 2\lambda^2 + 15\lambda - 14. \end{aligned}$$

Using Newton-Raphson method, we find that the eigenvalues of B are -3.47531 , 0.87584 & 4.59947 .
These are also the eigenvalues of A .

② Transform the matrix $A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 1 \end{bmatrix}$ to the

tridiagonal form by Givens method and hence find the eigenvalues of A .

Solⁿ:- let the orthogonal matrix S_1 be

$$S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

where $\tan\theta = \frac{a_{13}}{a_{12}} = 2$ i.e. $\theta = 1.1071$

$$\text{Therefore, } S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{5} & -2/\sqrt{5} \\ 0 & +2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix}$$

The reduced matrix is $B = S_1^{-1} A S_1$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{5} & 2/\sqrt{5} \\ 0 & -2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{5} & -2/\sqrt{5} \\ 0 & 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2\sqrt{5} & 0 \\ 2\sqrt{5} & 2.6 & -1.2 \\ 0 & -1.2 & -0.6 \end{bmatrix}$$

The Sturm sequence is:-

$$P_0(\lambda) = 1$$

$$P_1(\lambda) = 1 - \lambda$$

$$\begin{aligned} P_2(\lambda) &= (2.6 - \lambda)(P_1(\lambda)) - 20(P_0(\lambda)) \\ &= (2.6 - \lambda)(1 - \lambda) - 20 \\ &= \lambda^2 - 3.6\lambda - 17.4 \end{aligned}$$

$$\begin{aligned} P_3(\lambda) &= (-0.6 - \lambda)(P_2(\lambda)) - (1.2)^2(P_1(\lambda)) \\ &= (-0.6 - \lambda)(\lambda^2 - 3.6\lambda - 17.4) - 1.44(1 - \lambda) \\ &= \lambda^3 - 3\lambda^2 - 21\lambda - 9 \end{aligned}$$

Using Newton-Raphson method, we find that the eigenvalues of B are 6.46, -3 and -0.4641. These are also the eigenvalues of A.

③ Transform the matrix $M = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & -1 \\ 3 & -1 & 1 \end{bmatrix}$

to tridiagonal form by Givens method.

Soln: Let the orthogonal matrix S_1 be

$$S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

where $\tan\theta = \frac{a_{13}}{a_{12}} = \frac{3}{2}$

$$\cos\theta = \frac{2}{\sqrt{13}} \quad ; \quad \sin\theta = \frac{3}{\sqrt{13}}$$

$$\therefore S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{2}{\sqrt{13}} & -\frac{3}{\sqrt{13}} \\ 0 & \frac{3}{\sqrt{13}} & \frac{2}{\sqrt{13}} \end{bmatrix}$$

Thus, $B = S_1^{-1} A S_1$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{2}{\sqrt{13}} & \frac{3}{\sqrt{13}} \\ 0 & -\frac{3}{\sqrt{13}} & \frac{2}{\sqrt{13}} \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & -1 \\ 3 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{2}{\sqrt{13}} & -\frac{3}{\sqrt{13}} \\ 0 & \frac{3}{\sqrt{13}} & \frac{2}{\sqrt{13}} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & \sqrt{13} & 0 \\ \sqrt{13} & \frac{1}{13} & \frac{5}{13} \\ 0 & \frac{5}{13} & \frac{25}{13} \end{bmatrix} \quad \text{is the required tridiagonal form.}$$

Householder's method for a Symmetric matrix to reduce it into a Tridiagonal matrix :-

Procedure of the algorithm :- (for a 4×4 matrix)

First rotation :-

$$\text{let } V_2 = (0, s_2, s_3, s_4)^T$$

$$\text{and } q = \sqrt{a_{12}^2 + a_{13}^2 + a_{14}^2}$$

$$\text{where } s_2 = \frac{1}{2} \left(1 + \frac{a_{12} \times \text{sign}(a_{12})}{q} \right)$$

$$s_3 = \frac{a_{13} \times \text{sign}(a_{12})}{2q \times s_2}$$

$$s_4 = \frac{a_{14} \times \text{sign}(a_{12})}{2q \times s_2}$$

$$\text{Then, } S_2 = I - 2V_2 V_2^T$$

$$\text{and } A_2 = S_2 A_1 S_2$$

Second rotation :-

$$\text{let } V_3 = (0, 0, s_3, s_4)^T$$

$$\text{and } q = \sqrt{a_{23}^2 + a_{24}^2}$$

$$\text{where } s_3^2 = \frac{1}{2} \left(1 + \frac{a_{23} \times \text{sign}(a_{23})}{q} \right)$$

$$s_4 = \frac{a_{24} \times \text{sign}(a_{23})}{2q s_3}$$

$$\text{Then, } S_3 = I - 2V_3V_3^T$$

$$\text{and } A_3 = S_3 A_2 S_3$$

Procedure of the algorithm :- (for a 3×3 matrix)

First rotation:

$$\text{let } V_2 = (0, s_2, s_3)^T \text{ and } q = \sqrt{a_{12}^2 + a_{13}^2}$$

$$s_2^2 = \frac{1}{2} \left(1 + \frac{a_{12} \times \text{sign}(a_{12})}{q} \right)$$

$$s_3 = \frac{a_{13} \times \text{sign}(a_{12})}{2q s_2}$$

$$\text{Then, } S_2 = I - 2V_2V_2^T \text{ and } A_2 = S_2 A_1 S_2$$

① Use the Householder method to reduce the matrix

$$A = \begin{bmatrix} 2 & -1 & -1 & 1 \\ -1 & 4 & 1 & -1 \\ -1 & 1 & 3 & -1 \\ 1 & -1 & -1 & 2 \end{bmatrix} \text{ into the tridiagonal form}$$

Soln:- First rotation:-

$$\text{let } v_2 = (0, s_2, s_3, s_4)^T$$

$$\text{and } q = \sqrt{a_{12}^2 + a_{13}^2 + a_{14}^2}$$

$$= \sqrt{(-1)^2 + (-1)^2 + 1^2} = \sqrt{3}$$

$$\text{W.K.T } s_2^2 = \frac{1}{2} \left(1 + \frac{a_{12} \times \text{sign}(a_{12})}{q} \right)$$

$$= \frac{1}{2} \left(1 + \frac{(-1) \times (-)}{\sqrt{3}} \right) = 0.78868$$

$$\therefore s_2 = 0.88807$$

$$\text{Next, } s_3 = \frac{a_{13} \times \text{sign}(a_{12})}{2q \times s_2}$$

$$= \frac{(-1) \times (-)}{2\sqrt{3} \times 0.88807} = 0.32506$$

$$\text{Finally, } S_4 = \frac{1 \times (-1)}{2\sqrt{3} \times 0.88807} = -0.32506.$$

$$\text{Thus, } V_2 = (0, 0.88807, 0.32506, -0.32506)^T$$

$$\therefore S_2 = I - 2V_2V_2^T$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -0.57734 & -0.57735 & 0.57735 \\ 0 & -0.57735 & 0.78867 & 0.21133 \\ 0 & 0.57735 & 0.21133 & 0.78867 \end{bmatrix}$$

$$A_2 = S_2 A_1 S_2$$

$$= \begin{bmatrix} 2 & 1.73204 & 0 & 0 \\ 1.73204 & 5.0 & 0.21132 & -0.78867 \\ 0 & 0.21132 & 2.28867 & -0.5 \\ 0 & -0.78867 & -0.5 & 1.71133 \end{bmatrix}$$

Second rotation :-

$$V_3 = (0, 0, S_3, S_4)^T$$

$$q = \sqrt{a_{23}^2 + a_{24}^2}$$

$$= 0.81649$$

$$s_3^2 = \frac{1}{2} \left(1 + \frac{a_{23} \times \text{sign}(a_{23})}{q} \right) = 0.62941$$

$$\therefore s_3 = 0.79335$$

$$s_4 = \frac{a_{24} \times \text{sign}(a_{23})}{2q s_3} = -0.60876$$

$$\therefore V_3 = (0, 0, 0.79335, -0.60876)^T$$

$$S_3 = I - 2V_3V_3^T$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -0.25881 & 0.96592 \\ 0 & 0 & 0.96592 & 0.25882 \end{bmatrix}$$

$$A_3 = S_3 A_2 S_3$$

$$= \begin{bmatrix} 2 & 1.73204 & 0 & 0 \\ 1.73204 & 5.0 & -0.81648 & 0 \\ 0 & -0.81648 & 2 & -0.57731 \\ 0 & 0 & -0.57731 & 2 \end{bmatrix}$$

This is the required tri-diagonal matrix similar to A.

② Using the Householder's transformation reduce the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ into a tridiagonal matrix

Soln: Let $V_2 = (0, s_2, s_3)^T$

and $q = \sqrt{a_{12}^2 + a_{13}^2} = \sqrt{2}$

W.K.T $s_2^2 = \frac{1}{2} \left(1 + \frac{a_{12} \times \text{sign}(a_{12})}{q} \right)$

$$= \frac{1}{2} \left(1 + \frac{1}{\sqrt{2}} \right) = \frac{\sqrt{2} + 1}{2\sqrt{2}} = 0.8546$$

$\therefore s_2 = 0.9244$

$$s_3 = \frac{a_{13} \times \text{sign}(a_{13})}{2q s_2}$$

$$= \frac{1}{2 \times \sqrt{2} \times 0.9244} = 0.3836$$

Thus, $V_2 = (0, 0.9244, 0.3836)^T$

Now, $S_2 = I - 2V_2 V_2^T$

$$S_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -0.7090 & -0.7092 \\ 0 & -0.7092 & 0.7057 \end{bmatrix}$$

$$A_2 = S_2 A_1 S_2$$

$$= \begin{bmatrix} 2 & -1.41 & 0 \\ -1.41 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

This is the required tridiagonal matrix

Rutishauser method for arbitrary matrices.

Let A be a square matrix. In this method, a convergent sequence of upper triangular matrices $A_1, A_2, A_3, \dots$ are generated. The diagonal elements of the convergent matrix are the eigenvalues, if they are real.

The conversion is based on LU decomposition technique, where L and U are lower and upper triangular matrices. Initially let $A = A_1$ and $A_1 = L_1 U_1$ with $l_{ii} = 1$. Then form $A_2 = U_1 L_1$. The matrices A_1 and A_2 are similar as $A_2 = U_1 L_1 = U_1 A_1 U_1^{-1}$ and they have same eigenvalues.

Now, A_2 is factorized in the form of $A_2 = L_2 U_2$ with $l_{ii} = 1$. Then form $A_3 = U_2 L_2$. Proceeding this way we generate a sequence of similar matrices $A_1, A_2, A_3, \dots$. In general, this sequence converge to an upper triangular matrix. If the eigenvalues are real, then they all lie on the leading diagonal of the matrix obtained.

① Find all the eigenvalues of the matrix $\begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$ using Rutishauser method.

Solⁿ:- Let $A_1 = L_1 U_1$

$$\text{where } L_1 = \begin{bmatrix} 1 & 0 \\ l_{21} & 1 \end{bmatrix} \text{ and } U_1 = \begin{bmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{bmatrix}$$

$$\text{i.e. } \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} \\ u_{11}l_{21} & u_{12}l_{21} + u_{22} \end{bmatrix}$$

This gives $u_{11} = 4$, $u_{12} = 2$, $l_{21} = -\frac{1}{4}$, $u_{22} = \frac{3}{2}$

$$\text{Therefore, } L_1 = \begin{bmatrix} 1 & 0 \\ -\frac{1}{4} & 1 \end{bmatrix} \text{ and } U_1 = \begin{bmatrix} 4 & 2 \\ 0 & \frac{3}{2} \end{bmatrix}$$

$$\text{Form } A_2 = U_1 L_1 = \begin{bmatrix} 4 & 2 \\ 0 & \frac{3}{2} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -\frac{1}{4} & 1 \end{bmatrix} = \begin{bmatrix} \frac{7}{2} & 2 \\ -\frac{3}{8} & \frac{3}{2} \end{bmatrix}$$

$$\text{Again, let } A_2 = L_2 U_2 = \begin{bmatrix} 1 & 0 \\ l_{21} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{bmatrix}$$
$$\begin{bmatrix} \frac{7}{2} & 2 \\ -\frac{3}{8} & \frac{3}{2} \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} \\ u_{11}l_{21} & u_{12}l_{21} + u_{22} \end{bmatrix}$$

This gives $u_{11} = 7/2$, $u_{12} = 2$, $l_{21} = -\frac{3}{28}$, $u_{22} = \frac{12}{7}$

$$\text{Therefore, } L_2 = \begin{bmatrix} 1 & 0 \\ -\frac{3}{28} & 1 \end{bmatrix}, \quad U_2 = \begin{bmatrix} 7/2 & 2 \\ 0 & 12/7 \end{bmatrix}$$

$$\text{Form } A_3 = U_2 L_2 = \begin{bmatrix} 7/2 & 2 \\ 0 & 12/7 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -\frac{3}{28} & 1 \end{bmatrix} = \begin{bmatrix} \frac{23}{7} & 2 \\ -\frac{9}{49} & \frac{12}{7} \end{bmatrix}$$

In this way, we find

$$A_4 = \begin{bmatrix} 3.17391 & 2 \\ -0.10208 & 1.82609 \end{bmatrix}, \quad A_5 = \begin{bmatrix} 3.10959 & 2 \\ -0.06080 & 1.89041 \end{bmatrix}$$

$$A_6 = \begin{bmatrix} 3.07049 & 2 \\ -0.03772 & 1.92951 \end{bmatrix}, \quad A_7 = \begin{bmatrix} 3.04591 & 2 \\ -0.02402 & 1.95409 \end{bmatrix}$$

$$A_8 = \begin{bmatrix} 3.04591 & 2 \\ -0.00788 & 1.96986 \end{bmatrix}, \quad A_9 = \begin{bmatrix} 3.04073 & 2 \\ -0.00512 & 1.97504 \end{bmatrix}$$

The sequence $\{A_i\}$ converges slowly to an upper triangular matrix and the diagonal elements converge to the eigenvalues of A . The exact eigenvalues are 3 and 2, which are approximated by the diagonal elements of A_9 .

② Find all the eigenvalues of the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 1 & 3 & 2 \end{bmatrix}$$

using the Rutishauser method. Iterate till the elements of the lower triangular part are less than 0.05 in magnitude.

Soln:- We have

$$A_1 = A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = L_1 U_1$$

$$A_2 = U_1 L_1 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -1 & 1 \\ -2 & -1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ -1/2 & 1 & 0 \\ 1/4 & 7/6 & 1 \end{bmatrix} \begin{bmatrix} 4 & -1 & 1 \\ 0 & -3/2 & 1/2 \\ 0 & 0 & 1/6 \end{bmatrix} = L_2 U_2$$

$$A_3 = U_2 L_2 = \begin{bmatrix} 19/4 & 1/6 & 1 \\ 7/8 & -11/12 & 1/2 \\ 1/24 & 7/36 & 1/6 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 7/38 & 1 & 0 \\ 1/114 & -1/54 & 1 \end{bmatrix} \begin{bmatrix} 19/4 & 1/6 & 1 \\ 0 & -18/19 & 6/19 \\ 0 & 0 & 38/171 \end{bmatrix} = L_3 U_3$$

$$A_4 = U_3 L_3 = \begin{bmatrix} 4.789474 & -0.037037 & 1 \\ -0.171745 & -1.011696 & 0.315789 \\ 0.001949 & -0.045267 & 0.222222 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ -0.035859 & 1 & 0 \\ 0.000407 & 0.044670 & 1 \end{bmatrix} \begin{bmatrix} 4.789474 & -0.0370 & 1 \\ 0 & -1.0130 & 0.351 \\ 0 & 0 & 0.206 \end{bmatrix}$$

$$= L_4 U_4$$

$$A_5 = U_4 L_4$$

$$= \begin{bmatrix} 4.791209 & 0.007633 & 1 \\ 0.036469 & -0.997316 & 0.351648 \\ 0.000084 & 0.009207 & 0.206107 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0.007612 & 1 & 0 \\ 0.000018 & -0.009231 & 1 \end{bmatrix} \begin{bmatrix} 4.791209 & 0.0076 & 1 \\ 0 & -0.9973 & 0.344 \\ 0 & 0 & 0.209 \end{bmatrix}$$

$$= L_5 U_5$$

$$A_6 = U_5 L_5 = \begin{bmatrix} 4.791285 & -0.001598 & 1 \\ -0.007586 & -1.000550 & 0.344036 \\ 0.000004 & -0.001932 & 0.209265 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ -0.001583 & 1 & 0 \\ 0.000001 & 0.001931 & 1 \end{bmatrix} \begin{bmatrix} 4.7912 & -0.0015 & 1 \\ 0 & -1.0005 & 0.3456 \\ 0 & 0 & 0.2085 \end{bmatrix}$$

$$= L_6 U_6$$

$$A_7 = U_6 L_6$$

$$= \begin{bmatrix} 4.791289 & 0.000333 & 1 \\ 0.001584 & -0.999886 & 0.345619 \\ 0 & 0.000403 & 0.208597 \end{bmatrix}$$

Hence, the eigen values are approximately
 4.791289 , -0.999886 and 0.208597 .

Exercise :-

① Find all the eigenvalues of the matrix $A = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$
 using Rutishauser method.

Ans:-

The eigenvalues are 5 and 1.

Exercise :-

- ① Find the eigenvalues and eigenvectors of the symmetric matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ using Jacobi's method.

Ans:- Eigenvalues: 5, -1, -1 and the corresponding eigenvectors respectively are: $(0.57735, 0.57735, 0.57735)^T$, $(-0.70711, -0.70711, 0)^T$ and $(-0.40825, -0.40825, 0.81650)^T$.

- ② Find the eigenvalues of the following tri-diagonal matrix $A = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ using Givens method.

Ans:- 2, 0.26795 and 3.73205.

- ③ Use the Householder's method to reduce the given matrix A into the tridiagonal form

$$A = \begin{bmatrix} 4 & -1 & -2 & 2 \\ -1 & 4 & -1 & -2 \\ -2 & -1 & 4 & -1 \\ 2 & -2 & -1 & 4 \end{bmatrix}$$

Ans:-

$$\begin{bmatrix} 4 & 3 & 0 & 0 \\ 3 & 16/3 & -5/3\sqrt{5} & 0 \\ 0 & -5/3\sqrt{5} & 16/3 & 9/5 \\ 0 & 0 & 9/5 & 12/5 \end{bmatrix}$$